

On a population model with memory

Jean BERTOIN

University of Zürich

Memory in Evolutionary Biology

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Atavism: “A morphophysiological trait controlled by an ancient gene regulatory network that survives periods of disuse and can be reactivated and reused in later lineages, even if it was not used in the immediate ancestors” (G. Wagner)

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Actually, it is always beneficial to use the memory from preceding generations for reproduction.

Population model with no memory

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Basic reproduction kernel

$$\pi(s, n), \quad s \in S \text{ and } n \in \mathbb{N}^S,$$

gives the probability that a parent with type s begets exactly n_t children of type t for every $t \in S$.

Population model with no memory

The large time behavior of a multitype branching process is determined by spectral properties of the **mean reproduction matrix**

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The mean reproduction m defines a square matrix with nonnegative entries.

We assume that is **irreducible**.

Spectral radius

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determines the growth rate of the population (on average)

$$m^n \mathbf{1} \sim r^n.$$

By the Perron–Frobenius theorem, r is a simple eigenvalue of m .

A left-eigenvector can be chosen to be a probability measure ρ on S

$$\rho m = r \rho.$$

The repartition of types is close to

$$r^{-n} m^n 1_s \sim \rho(s).$$

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At reproduction, a generation T is activated at random.

Given the type of the forebear T generations backwards, $s_T = s'$,
the statistics of the reproduction are $\pi(s', \cdot)$.

Population model with memory

A child born with type $t \in S$ from a parent with memory $\mathbf{s}$ has then memory

$$t\mathbf{s} = (t, s_0, s_1, \dots).$$

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Assume i.i.d. variables T with a fixed law τ are used to select the activated generation for different individuals.

Mean reproduction with memory

The population model with memory can be seen as a branching process with an uncountable space of types (space of memories)

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$\mathcal{S}$ is equipped with a natural distance, and is then compact.

The mean reproduction operator $\mathbf{m}$ acts on the space of bounded measurable functions $f : \mathcal{S} \rightarrow \mathbb{R}$,

$$\mathbf{m}f(\mathbf{s}) = \sum_{j=0}^{\infty} \tau(j) \sum_{t \in \mathcal{S}} m(s_j, t) f(t\mathbf{s}), \quad \mathbf{s} \in \mathcal{S}. \quad (2)$$

Lemma

- *The Feller property always holds,*

$$m(\mathcal{C}(S)) \subset \mathcal{C}(S).$$

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$$m(\mathcal{B}(S)) \not\subset \mathcal{C}(S).$$

- *The spectral radius r of m is a left-eigenvalue of m ,*

$$\rho m = r \rho$$

for some probability measure ρ on S .

Ruelle's extension of the Perron–Frobenius theorem implies that if

$$\text{all the entries } m(s, t) > 0,$$

and the memory has an exponential tail

$$P(T > n) = o(a^n) \quad \text{for some } a < 1,$$

then ρ is unique and furthermore there exists a unique continuous function $h : S \rightarrow (0, \infty)$ with

$$mh = rh \quad \text{and} \quad \rho(h) = 1.$$

Theorem

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Interpretation: The average growth of the population with memory is always faster than for the population without memory.

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$$\bar{r} \geq r.$$

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When the memory law τ has a finite expectation, the inequality is strict except if either $s \mapsto \sum_t m(s, t)$ or T are constant.

An example

$$S = \{a, b\}, m = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The spectral radius of m is $r = (3 + \sqrt{5})/2 \approx 2.618$.

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Four memory sequences of length 2 in the lexicographic order, (a, a) , (a, b) , (b, a) and (b, b) . Then

$$m = m(u) = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 1+u & 0 \\ 0 & 1 & 0 & 2-u \\ 0 & 1 & 0 & 2 \end{pmatrix}.$$

An example

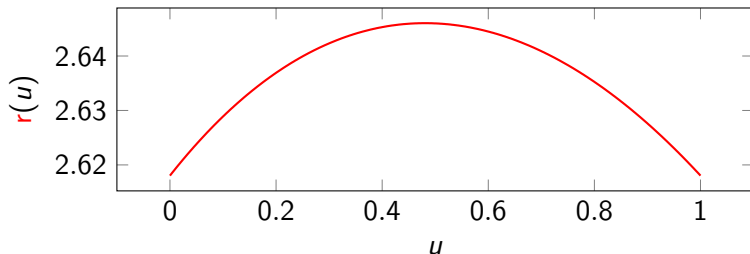
The characteristic polynomial of $\mathbf{m}(u)$ is

$$\chi_u(\lambda) = \lambda^4 - 3\lambda^3 + (1 - u)\lambda^2 + (2u + u^2)\lambda - u^2$$

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Plot of the spectral radius $r(u)$ of $\mathbf{m}(u)$

Some elements of the proof

The proof uses a **many-to-one formula**.

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There is a great variety of such formulas, and our choice is guided by the spectral properties of the model without memory.

A biased Markov chain

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Recall that $r > 0$ denotes the spectral radius of the mean reproduction matrix m and from the Perron–Frobenius theorem that there is a unique solution (ρ, h) of the eigen-problem

$$\rho m = r\rho \quad \text{and} \quad mh = rh,$$

where ρ is a probability measure on S and $h : S \rightarrow (0, \infty)$ a function with

$$\rho(h) = \sum_{s \in S} \rho(s)h(s) = 1.$$

A biased Markov chain

This incites us to introduce the **normalized** transition matrix $\overline{m}$

$$\overline{m}(s, t) = \frac{m(s, t)h(t)}{rh(s)}, \quad s, t \in S, \quad (3)$$

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$$\bar{m}(s, t) = \frac{m(s, t)h(t)}{rh(s)}, \quad s, t \in S, \quad (3)$$

and then the **Markovian** transition kernel

$$\bar{Q}f(s, \textcolor{red}{s}) = \sum_{j=0}^{\infty} \tau(j) \sum_{t \in S} \bar{m}(s_j, t) f(s_j, t\textcolor{red}{s}), \quad f \in \mathcal{B}(\bar{S}), \quad (4)$$

where $\bar{S} = S \times \textcolor{red}{S}$.

A many-to-one formula

We write $\overline{X} = ((Y_k, X_k) : k \geq 0)$ for the Markov chain in $\overline{S}$ with transition kernel $\overline{Q}$ and call it the biased chain.

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In this framework, the many-to-one formula reads

$$m^k f(s) = r^k \overline{E}_s \left(f(X_k) \times \prod_{j=1}^k \frac{h(Y_j)}{h(X_j)} \right), \quad f \in \mathcal{B}(S), \quad (5)$$

where $X_j = (X_j, X_{j+1}, \dots)$.

A many-to-one formula

Hence the asymptotic behavior of m^k can be investigated from that of the weights

$$\prod_{j=1}^k \frac{h(Y_j)}{h(X_j)} = \exp \left(\sum_{j=1}^k \log h(Y_j) - \sum_{j=1}^k \log h(X_j) \right).$$

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This leads us to analyze ergodic properties of the biased chain.

Invariant distribution of the biased chain

The biased Markov chain $\overline{X} = ((Y_k, X_k) : k \geq 0)$ possesses at least one invariant distribution $\overline{\sigma}$,

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(Feller on a compact state space).

The one-dimensional marginals of $\overline{\sigma}$ are given explicitly in terms of the eigen-elements h and ρ :

$$\overline{\sigma}(\{(s, \mathbf{s}) : s = t\}) = \overline{\sigma}(\{(s, \mathbf{s}) : s_j = t\}) = \rho(t)h(t).$$

Invariant distribution of the biased chain

In particular, one has

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As a consequence, by Jensen's inequality, for any $k \geq 1$,

$$\overline{E}_{\overline{\sigma}}\left(\prod_{j=1}^k \frac{h(Y_j)}{h(X_j)}\right) \geq 1.$$

The many-to-one formula then yields

$$\sigma \mathbf{m}^k \mathbf{1} \geq r^k,$$

which in turn forces $r \geq r$ (by Gelfand's formula).

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Open questions

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- How does memory affect the asymptotic repartition of types ?
- Is there always a **unique** probability law ρ on S with $\rho m = r \rho$?
- Is there always a function $h : S \rightarrow (0, \infty)$ with $mh = rh$?